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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: APPLIED MACHINE LEARNING FOR TEXT ANALYSIS

Course code: 24ALT3101

A.Y. 2026 – 2027

TITLE OF THE PROJECT	**Deep Learning Framework for Automated Pneumonia Detection Using Chest X-ray Images**	
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Abstract

Pneumonia is one of the leading causes of death worldwide and requires early diagnosis for effective treatment. Chest X-ray imaging is the most widely used diagnostic technique; however, manual interpretation by radiologists is time-consuming and prone to errors, particularly in hospitals with limited medical resources. This project proposes an Explainable Deep Learning Framework for Automated Pneumonia Detection Using Chest X-ray Images by integrating Convolutional Neural Networks (CNN), DenseNet-121, EfficientNet, Transformer models, and Bidirectional Encoder Representations from Transformers (BERT) to improve diagnostic accuracy and interpretability. The proposed framework utilizes multiple publicly available Chest X-ray datasets to improve model robustness and generalization across diverse patient populations. The system performs image preprocessing, augmentation, feature extraction, deep learning-based classification, and prediction through a comprehensive AI pipeline. DenseNet and EfficientNet are employed for extracting high-level visual features, while Transformer architectures enhance feature representation and contextual understanding. Explainable AI techniques such as Grad-CAM are incorporated to highlight infected lung regions, enabling clinicians to understand the reasoning behind model predictions. Furthermore, BERT combined with a Large Language Model (LLM) generates simplified diagnostic summaries and clinical explanations based on the predicted results, making the system more accessible to healthcare professionals and patients. A complete web-based application consisting of a React frontend and FastAPI backend is developed to provide real-time image upload, pneumonia prediction, confidence scores, heatmap visualization, AI-generated medical reports, and patient history management. The proposed framework is evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix to ensure reliable performance. The developed system aims to support hospitals and healthcare professionals by providing an accurate, scalable, explainable, and AI-assisted pneumonia detection solution.

Dataset	URL link
NIH ChestX-ray14	https://www.kaggle.com/datasets/nih-chest-xrays/data
2. RSNA Pneumonia Detection Challenge	https://www.kaggle.com/c/rsna-pneumonia-detection-challenge
3. VinDr-CXR	https://github.com/t-davidson/hate-speech-and-offensive-language

Approach:

Multiple Public Chest X-ray Datasets

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Dataset Cleaning & Label Mapping

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Image Preprocessing
- Resize Images
- Normalize Pixel Values
- Data Augmentation
- Remove Corrupted Images

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Train-Test Split (80:20)

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Image Feature Extraction
(CNN / DenseNet / EfficientNet)

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Transformer-based Feature Learning

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Dense Classification Layer

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Pneumonia / Normal Prediction

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Explainable AI (Grad-CAM)

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BERT + Large Language Model
AI Medical Report Generation

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FastAPI Backend

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React Frontend Dashboard

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Patient Report & Prediction History

Signature of the Guide HOD